LLM Baseline Experiment

Project: AI-Assisted Travel Planning Web Application

1. Experiment Purpose

This preliminary experiment examines how existing general-purpose LLM systems respond to the same travel-planning request. The aim is to establish a baseline for what users can already obtain directly from current AI systems and to identify limitations that the proposed application may address.

2. Systems Tested

- DeepSeek
- Claude
- Gemini

The same travel-planning prompt was used for all three systems to make the comparison as consistent as possible.

3. Test Scenario

- Route: Edinburgh to London.
- Trip type: short solo leisure trip.
- Duration: 2 days.
- Total budget: £500, including transport, accommodation, food and attractions.
- Interests: museums and history, local food, photography and interesting neighbourhoods.
- Required output: transport, accommodation, morning/afternoon/evening plans, food recommendations, approximate costs, total estimated budget and explanations of why activities match the stated interests.
- The models were also asked to identify uncertain or changeable information rather than inventing precise details.

4. Evaluation Criteria

- Completeness - whether all requested parts of the itinerary were included.
- Personalisation - whether recommendations reflected the stated interests.
- Budget handling - whether the plan considered the £500 constraint and whether estimates appeared plausible.
- Temporal consistency - whether travel time, activity timing and number of nights were logically consistent.
- Uncertainty handling - whether changing prices, opening hours or other dynamic information were clearly identified.
- Structure and usability - whether the output was easy to read and act on.

5. Results

DeepSeek

DeepSeek produced a complete and well-structured itinerary with transport, accommodation, daily activities, food suggestions and a budget summary. It matched activities to the user interests effectively.

However, it made stronger assumptions about missing information. In particular, it treated a two-day trip as two nights of accommodation and did not fully integrate the Edinburgh-to-London travel time into the first morning schedule. Its total estimated cost of £185 also appeared optimistic and would require further verification.

Claude

Claude produced a detailed itinerary with explicit travel timing, one overnight stay and a clearer explanation of uncertain information. It used wider cost ranges and highlighted factors such as changing train fares, hotel prices, opening hours and railway engineering works.

The response showed stronger temporal consistency and more cautious budget handling, but it was considerably longer and more information-dense than the other outputs, which may make it harder for users to scan quickly.

Gemini

Gemini produced a compact and structured itinerary with place information, estimated costs and a total budget of approximately £400. It used one overnight stay and presented the information in a product-like format.

A consistency issue was observed where venue availability information appeared alongside an itinerary that still scheduled the venue. This suggests that access to dynamic information does not automatically guarantee that the generated plan uses that information correctly as a constraint.

6. Comparison Summary

Criterion	DeepSeek	Claude	Gemini
Completeness	High	High	High
Personalisation	Good	Good	Good
Budget handling	Optimistic	Cautious	Moderate
Temporal consistency	Some issues	Strong	Moderate
Uncertainty handling	Limited	Strong	Moderate
Presentation	Clear	Detailed but long	Compact / product-like

7. Main Findings

- All three systems were already capable of generating useful personalised travel itineraries.
- The main limitations were not basic itinerary generation, but assumptions, price realism, timing consistency and the handling of dynamic information.
- A generated plan may look plausible while still containing information that requires verification.
- Long chatbot responses can also make useful information harder to scan quickly.

8. Implications for the Proposed System

The experiment indicates that the proposed system should not focus only on generating an itinerary. Its value should come from improving how AI-generated plans are collected, checked and presented.

- Use structured user inputs to reduce hidden assumptions.
- Show budget information clearly and support basic constraint checking.
- Consider travel-time and schedule consistency.
- Mark prices, opening hours and other dynamic information as estimated or requiring verification.
- Present the itinerary in a concise day-by-day interface instead of a long chatbot-style answer.

9. Preliminary Conclusion

The baseline experiment shows that general-purpose LLMs can already generate strong travel suggestions. The proposed application therefore needs to add value through structured input, clearer presentation, basic validation and more transparent handling of uncertain information. These findings will guide the next stage of system design and implementation.